

# Social Media Usage and Self-Reported Mental Health

A Behavioral Classification Study

Course: 36103 Statistical Thinking for Data Science — 2026

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# Executive Summary

This study analysed behavioural data from 23,678 social media users to identify factors related to negative self-reported mental health. Comparing Decision Tree, Random Forest, Logistic Regression and KNN across three feature sets, daily screen time and addiction level were the strongest predictors. Users spending more than 150 minutes per day were more likely to report negative outcomes, with sleep duration as a moderator. Platform type and purpose of use showed limited predictive value. As the data is synthetic, findings are directional. We recommend digital-wellness tools centred on screen-time limits and sleep monitoring.

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# 1. Introduction

Social media is central to everyday life, with billions of users spending several hours a day on platforms like TikTok, Instagram, and YouTube. Growing evidence suggests heavy use comes at a cost to well-being: it is linked to social isolation as online interactions displace face-to-face connection (Primack et al., 2017), to body-image concerns, eating disorders and suicidal ideation among young people (Suhag & Rauniyar, 2024), and to poorer sleep via bedtime cognitive arousal and late-night scrolling driven by fear of missing out (Kinsella & Chin, 2024).

Using a dataset of 23,678 social media users across 45 variables, this study investigates: Is social media usage intensity associated with a negative impact on mental well-being, and what behavioural factors drive this relationship?

Three sub-research questions guide our analysis:

SRQ1 — Are screen time and addiction level associated with self-reported mental health?

SRQ2 — Is usage intensity associated with shorter sleep? SRQ3 — Do platform type and purpose add predictive value beyond behavioural variables?

## 2. Dataset and Data Cleaning

The dataset was sourced from the Kaggle 2026 Global Social Media Users Dataset (Khurram, 2026): 25,000 synthetic user records across 45 behavioural, demographic, and platform-related variables. As the data is synthetic, findings should be interpreted cautiously.

Cleaning focused on real-world plausibility. Records were removed for users under 18 recorded as married, widowed or divorced; under 25 with a Master's or under 15 with a Bachelor's; under 16 in qualified occupations; and users whose account age exceeded their actual age. In total, 1,322 records (5.3%) were removed, leaving 23,678 observations with no missing values. Two inconsistencies were retained: non-zero spending despite non-purchaser flags, and 915 self-identified creators with zero posts (the latter fed a `creator_tier` feature). Skewed variables (followers, following, notifications, spending) were log-transformed; a `followers_velocity` feature was derived. The target was binarised into Negative vs Non-negative; class distribution (Figure 1) showed moderate imbalance (~31/69), addressed via `class_weight='balanced'`.

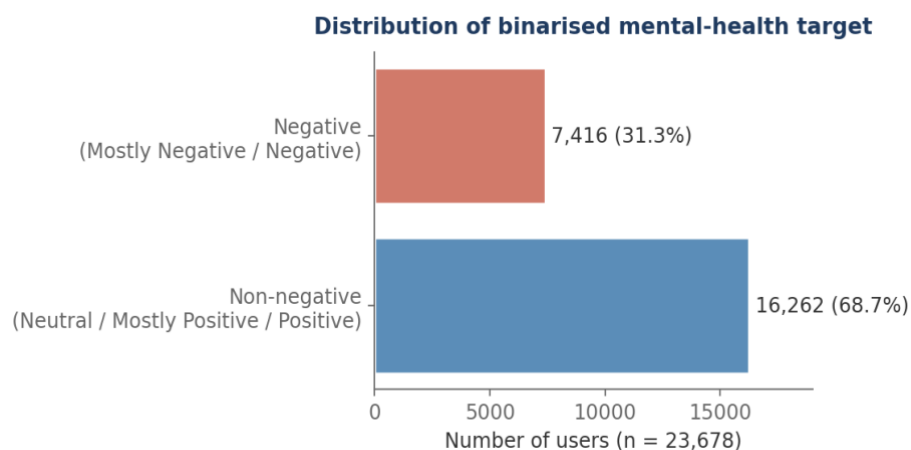


Figure 1. Distribution of the binarised mental-health target variable ( $n = 23,678$  users).

Exploratory analysis revealed that daily screen time, sleep hours, addiction level and age had the strongest univariate associations with the target, while platform-level variables and stated purpose showed weak correlations. This motivated three feature configurations: Behaviour Only; Behaviour + Key Categories (cyberbullying, screen-time-limit use, phone-checking habits, creator tier); and Behaviour + Rich Categories (platform, content type, occupation, relationship status), testing whether contextual variables add incremental value.

## 3. Modelling Strategy

### 3.1 Cross-Validation Strategy

We used cross-validation to evaluate model performance, repeatedly splitting the data into training and validation folds while preserving the original class proportions (~69% non-negative, 31% negative). Repeating this across random splits reduced the chance that results were driven by a single favourable partition. We also kept separate stratified train-test splits to produce confusion matrices on unseen data.

### 3.2 Evaluation Metrics

Three metrics were used:

- **F1 Score (primary):** balances precision and recall, more appropriate than accuracy under class imbalance.
- **ROC AUC:** measures how well each model separates negative from non-negative outcomes across thresholds.
- **Accuracy:** included for context only, since it can look artificially high when one class dominates.

Precision and recall were also examined to see how well each model identified the minority negative-impact group.

### 3.3 Models and Feature Sets

We compared four algorithms rather than relying on one: Logistic Regression (parametric baseline with interpretable coefficients), Decision Tree (intuitive rules, captures non-linearities), Random Forest (ensemble of trees, more stable and better at variable interactions), and KNN (distance-based, classifies by similarity rather than learned rules). This let us balance predictive performance with interpretability.

To handle class imbalance, balanced class weights were applied to Logistic Regression, Decision Tree, and Random Forest. Since Logistic Regression and KNN are sensitive to scale, numerical predictors were standardised.

Three feature configurations were compared, giving 12 models in total:

- **Behaviour Only:** numerical behavioural and engagement variables.
- **Behaviour + Key Categories:** behavioural variables plus high-impact categorical variables identified in EDA.
- **Behaviour + Rich Categories:** behavioural variables plus a broader set of demographic and categorical features.

This let us test whether adding contextual or demographic information improved performance beyond behavioural variables alone.

## 4. Results

### 4.1 Model Comparaison

All 12 models were compared using F1 Score and ROC AUC (Table 1). Random Forest and Decision Tree performed best with effectively identical results ( $F1 \approx 0.60$ ,  $ROC\ AUC \approx 0.76$ ). Logistic Regression was competitive with interpretable coefficients; KNN consistently underperformed.

*Table 1. Cross-validated performance of all 12 model  $\times$  feature-set combinations, sorted by F1 Score.*

| Model               | Feature Set           | Accuracy | F1 Score | ROC AUC |
|---------------------|-----------------------|----------|----------|---------|
| Random Forest       | Behaviour + Key Cat.  | 0.7183   | 0.6022   | 0.7572  |
| Decision Tree       | Behaviour + Rich Cat. | 0.7167   | 0.6020   | 0.7581  |
| Random Forest       | Behaviour Only        | 0.7179   | 0.6020   | 0.7579  |
| Decision Tree       | Behaviour Only        | 0.7168   | 0.6018   | 0.7590  |
| Decision Tree       | Behaviour + Key Cat.  | 0.7167   | 0.6017   | 0.7588  |
| Random Forest       | Behaviour + Rich Cat. | 0.7161   | 0.6004   | 0.7558  |
| Logistic Regression | Behaviour + Key Cat.  | 0.6942   | 0.5892   | 0.7540  |
| Logistic Regression | Behaviour Only        | 0.6934   | 0.5885   | 0.7541  |
| Logistic Regression | Behaviour + Rich Cat. | 0.6904   | 0.5839   | 0.7519  |
| KNN                 | Behaviour Only        | 0.7273   | 0.4806   | 0.7248  |
| KNN                 | Behaviour + Key Cat.  | 0.7229   | 0.4630   | 0.7104  |
| KNN                 | Behaviour + Rich Cat. | 0.6857   | 0.3112   | 0.6256  |

### 4.2 Selected Model: Decision Tree — Behaviour Only

Although Random Forest achieved a marginally higher F1 Score, the Decision Tree using the Behaviour Only feature set was selected as the final model due to its substantially greater interpretability and near-identical predictive performance. This made it more appropriate for communicating behavioural risk pathways to non-technical stakeholders.

The selected model also demonstrated that behavioural variables alone were capable of producing meaningful classification performance without requiring extensive demographic or platform-specific information.

### 4.3 Decision Tree Key Rules

The Decision Tree identified several interpretable behavioural pathways associated with elevated risk of negative mental health perception:

- Screen time above approximately 150 minutes per day consistently directed users toward higher-risk branches.
- Users under approximately 50 years old combined with high screen time showed an estimated 68% probability of belonging to the negative mental health class.
- Even among users with lower screen time, sleep duration below approximately 6.85 hours per night still frequently resulted in negative classifications, suggesting sleep acted as an independent moderating factor.

Together, these rules suggest that negative mental health outcomes were associated with combinations of behavioural intensity, age, and sleep-related behaviours rather than any single factor alone.

#### 4.4 Feature Importance

Feature importance analysis from the tree-based models consistently identified screen exposure, sleep behaviour, addiction level, and engagement intensity as the strongest predictors of negative mental health outcomes.

Daily screen time emerged as the most influential predictor across both Decision Tree and Random Forest models. Other highly important variables included sleep hours per night, addiction level, daily notifications, and engagement-related activity measures. These findings aligned closely with patterns previously identified during exploratory analysis.

Interestingly, richer demographic and platform-specific categorical variables contributed relatively limited additional predictive power compared with core behavioural metrics.

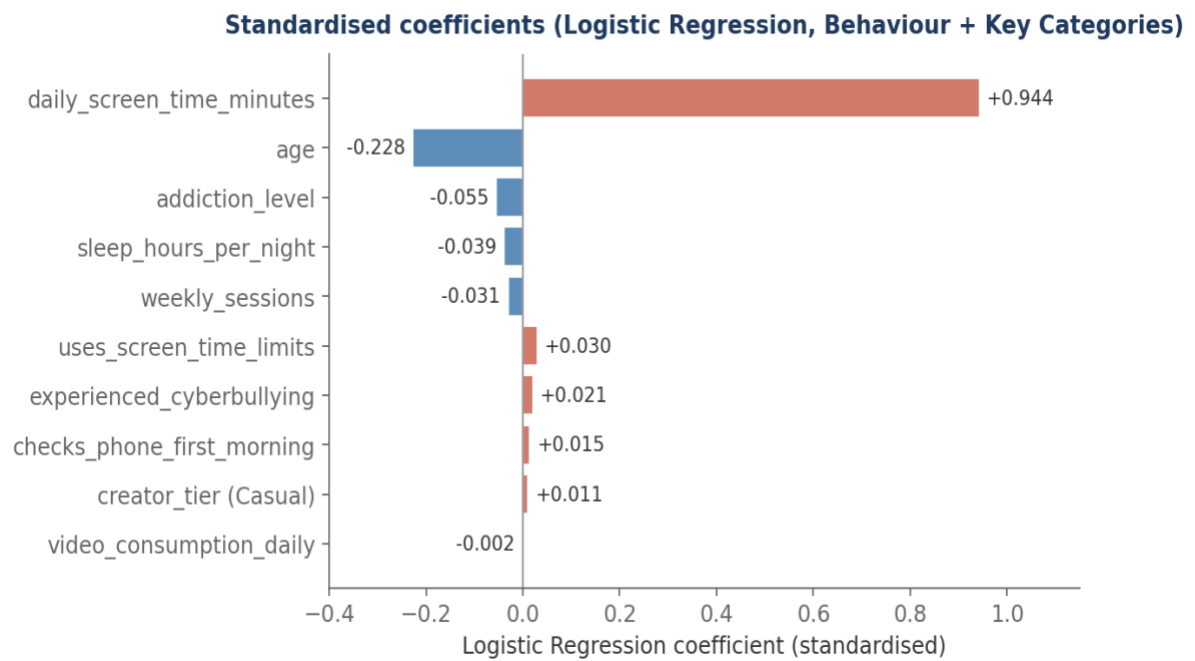
#### 4.5 Logistic Regression

Logistic Regression produced competitive predictive performance while also providing interpretable coefficients that helped quantify the direction and relative strength of predictors.

The strongest positive coefficient was observed for daily screen time (0.9435), indicating that higher daily screen exposure was strongly associated with increased likelihood of reporting negative mental health effects.

Age produced the strongest negative coefficient (-0.2277), suggesting that younger users were more likely to belong to the negative mental health class. Additional negative relationships were observed for sleep hours per night (-0.0393) and weekly sessions (-0.0312), supporting the broader finding that behavioural intensity and reduced sleep were closely linked to negative outcomes.

Among the categorical variables, cyberbullying experience, checking the phone first thing in the morning, and creator-tier status showed smaller positive associations with the negative mental health class, although their effects were substantially weaker than daily screen time.



*Figure 2. Standardised Logistic Regression coefficients (Behaviour + Key Categories). Red = positive association with negative mental health; blue = negative association.*

# 5. Discussion

## 5.1 Research Question answers

Screen time and addiction level were strongly linked to negative self-reported mental health, confirmed by both Logistic Regression and Decision Tree results. SRQ2 — Sleep hours acted as a moderator:  $< \sim 7$  hours linked to worse outcomes even among moderate users. SRQ3 — Platform type and usage purpose added little predictive value: usage intensity matters more than where or why. Findings align with Karim et al. (2020) and Kinsella & Chin (2024).

## 5.2 Limitations

Synthetic data limits generalisability; the self-reported target risks response bias; moderate F1 ( $\sim 0.60$ ) implies unmeasured confounders (pre-existing conditions, socioeconomic context); binarising the target lost ordinal detail; findings are correlational, not causal.

## 5.3 Ethical considerations

Mental health data is sensitive; real-world applications require informed consent, transparency, and strong privacy protections. When working with data related to Aboriginal and Torres Strait Islander peoples, research should follow Indigenous Data Sovereignty principles — collective consent, community partnership, community-led governance. The IGA module reframed our future practice that ethical considerations enter at dataset design, not deployment: co-designing variables with communities, keeping data under community governance, and recognising that statistical "objectivity" can mask cultural assumptions in the data. Culturally informed engagement is a precondition for analysis, not an add-on.

## 6. Conclusion

Across 23,678 users, higher daily screen time and addictive patterns predicted poorer self-reported mental well-being, with sleep moderating the effect; platform and stated purpose added little. We recommend evidence-based daily screen-time limits (e.g., 1–2 hours of non-essential use, stricter for adolescents) and sleep-aware nudges in digital-wellness tools. Future research should use longitudinal data with clinical mental-health measures to assess causality.

# References

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# Appendix A — Code

The complete reproducible analysis is provided in the accompanying Jupyter notebook `Assignment_2_Code.ipynb`. The notebook is self-contained and runs end-to-end on a Python 3 environment with the standard scientific stack (pandas, numpy, scikit-learn, statsmodels, matplotlib, seaborn).

The code file is organized as follow :

1. Data loading and inspection (cells 1–10) — loads the Kaggle CSV, converts `account_created_date` to datetime, generates summary statistics.
2. Data cleaning (cells 11–16) — applies four plausibility filters (age vs marital status, education timeline, occupation by age, account age vs user age), removing 1,322 rows.
3. Feature engineering (cells 17–22) — log-transforms skewed counts, derives `followers_velocity`, builds `creator_tier`, binarises target.
4. Exploratory analysis (cells 23–30) — univariate and bivariate plots for numerical and categorical features against the target.
5. Modelling (cells 31–40) — fits Logistic Regression, Decision Tree, Random Forest, and KNN across three feature sets using `RepeatedStratifiedKFold` cross-validation.
6. Evaluation (cell 41) — aggregates and compares all 12 models.